

Subject Description Form

Subject Code	AP20011
Subject Title	Optics
Credit Value	3
Level	2
Exclusion	AP20009 and AP20010
Objectives	The objectives of this subject are to provide students with fundamentals in geometric and wave optics for their future study of visual and ophthalmic science.
Intended Learning Outcomes	On completing the subject, students will be able to (a) describe the physical phenomena of practical geometrical and wave optics; (b) identify and solve some optical problems; and (c) demonstrate knowledge of geometrical and wave optics essential for the future study of visual and ophthalmic science.
Subject Synopsis/ Indicative Syllabus	Gaussian optics - reflection and refraction at plane and spherical surfaces; thin lens; cardinal points of thick lenses and lens systems; image forming systems. Aberrations - dispersive power of optical materials; achromatic prisms and lenses; monochromatic aberrations; ray tracing. Optical instrumentation: prism, stops, pupils and windows; simple magnifiers and eyepieces, telescopes. Wave nature of light - Polarization by reflection, dichroism and birefringent; polarizers and applications; Interference, interferometry and applications; optical coating; Fresnel and Fraunhofer diffraction; Resolving power; principle and applications of lasers. Production and Measurement of light - fundamental laws of photometry; Reflectance, transmittance and absorbance. Practical work: experiments on geometrical and wave optics.
Teaching/Learning Methodology (Note 3)	Lecture: The course contents will be delivered through lecture in class. Active participation in discussion by students will be encouraged. Tests will be given to class at appropriate intervals to consolidate students' understanding of the acquired knowledge as well as to enhance their problem-solving skills. Tutorial: Small group tutorials will be conducted to let the students have a better and deeper understanding on the course contents and improve their problem-solving skill. The students are encouraged to do a timely revision in order to solve the tutorial questions. Laboratory: Students will be given opportunities for hands-on experiments. They are expected to gain direct experience in using basic optical instruments and to apply their learnt knowledge for practical applications. Through the process students can enhance their analytical power, problem solving skills, conceptual understanding, critical thinking and desire for life-long learning.
Assessment Methods in Alignment with Intended Learning Outcomes	

	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
			a	b	c
	(1) Continuous assessment: tests, (practical) assignments	40	✓	✓	✓
	(2) Examination	60	✓	✓	✓
	Total	100			
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessment (assessment method 1), including homework problem sets, tests, lab report and a final written examination (assessment method 2), all require demonstration of understanding of the relevant course materials, good problem-solving skills, and being able to relate the acquired knowledge to solve practical optics problems. The continuous assessments are designed to monitor the study progress of the student. They also serve to provide student with a mechanism of self-evaluation of learning achievement. The final written examination will be used to assess the knowledge acquired by the students; as well as to determine the level of attainment of the prescribed learning outcomes.				
Student Study Effort Expected	Class contact:				
	▪ Lecture & Tutorial				26 h
	▪ Practical (labs)				20 h
	Other student study effort:				
	▪ Self-study				74 h
	Total student study effort				120 h
Reading List and References	Pedrotti, Frank L., Leno M. Pedrotti, and Leno S. Pedrotti. “Introduction to optics”, Cambridge University Press, 2017.				
	Tunnacliffe A.H., “Introduction to Visual Optics”, 4th edition Association of British Dispensing Opticians, 2010.				
	Pedrotti L.S. and Pedrotti F.L., “Optics & Vision”, Prentice-Hall, 1998.				